

Trades-Runtime

Workforce Capacity, Availability & Demand-Aware Scheduling - Addendum v0.8

Scope. This addendum contains all architecture added since Universal Analytics & Human Review Layer v0.7. It defines real-world employee availability constraints, demand-responsive booking blocks, scheduled-off-day calibration, probabilistic call duration, emergency/urgency handling, first-call and midday schedule protection, reschedule minimization and continuous capacity recalibration.

1. Workforce Availability & Capacity Constraint Engine

Trades-Runtime must schedule against the workforce that actually exists, not an imaginary fully staffed board. Employee availability, scheduled time off, call-ins, early-departure requirements, delayed starts, training, vehicle availability and authorized scheduling constraints become part of the canonical live company state.

Demand Forecast + Workforce Availability + Employee Windows + Skill Capacity + Existing Commitments + Runtime/Travel Prediction -> Live Capacity State -> Booking/Dispatch Blocks -> Continuous Recalibration -> Actual Outcome -> Calibration Learning.

2. Hard vs Soft Constraints

Constraint	Treatment
Hard availability	Approved PTO/day off, call-in, leave, must-leave-by, cannot-start-before, training, licensing restriction, unavailable Van o
Soft optimization	Preferences and operating choices such as preferred return time, protected emergency capacity, maintenance reduction

3. Employee Availability Windows

A time restriction is not merely a calendar label. Trades calculates backward from the employee's required end/return time using live location, predicted job duration, uncertainty, expected travel and destination.

Availability window -> predicted job duration -> uncertainty buffer -> travel/return time -> latest safe dispatch time.

Example: if a technician must leave by 3:30 PM and be back at the shop by 4:00 PM, Trades determines the latest feasible new assignment dynamically rather than simply marking 4:00 PM unavailable.

4. Demand-Responsive Booking Blocks

Trades can recommend, and where explicitly authorized can implement, booking-capacity blocks based on live and forecast demand. This includes reducing maintenance availability during severe service demand, employee call-ins, constrained skill mix or emergency surges.

Booking state	Meaning
OPEN	Available for normal booking.
HOLD	Temporarily reserved while demand/capacity is evaluated.
BLOCK NEW BOOKING	Do not add new routine work into the window.
PROTECTED	Reserved for defined capacity purpose such as emergencies or existing commitments.
RESCHEDULE CANDIDATE	Existing flexible commitment identified for human-reviewed displacement if required.
CLOSED	Unavailable because of hard workforce/operational constraint.

5. Existing Customer Commitments Remain Protected

Blocking capacity does not silently cancel already booked customers. Existing bookings retain precedence under the Trades customer-commitment rule. Displacement must be explicit, auditable, customer-impact aware and evaluated against alternatives. Human authority remains available for all material schedule changes.

6. Live Absence / Return Recalculation

When an employee calls in, Trades immediately removes the relevant labor/skill capacity, recalculates booked capacity and Van trajectories, identifies calls at risk, evaluates maintenance blocks, cross-branch borrowing and other recovery options, then presents ranked recovery plans. If the employee later becomes available, Trades restores capacity and recomputes from the actual current state.

Call-in -> capacity loss -> at-risk commitments -> recovery alternatives -> management decision -> live rebase -> settlement.

7. Real Usable Operating Capacity

Headcount alone is not capacity. Trades should calculate usable capacity from available labor, skill mix, Van availability, employee windows, call demand, expected runtime, geography, protected commitments, weather, inventory and absence state.

Available labor x skill mix x Van availability x time windows x call demand x expected runtime x geography x protected commitments x absence state x weather x inventory = real usable operating capacity.

8. Workforce Calibration

Historical availability events should be aggregated against demand and operational outcomes. Trades learns how specific combinations of absences, skill loss, weather, weekday, booked load and branch conditions affect customer wait, overtime, first-trip completion, displacement, maintenance blocking, cross-branch support and realized economics.

9. Demand-Aware Booking & Schedule Reliability Engine

Trades must optimize for service demand, urgency, employee availability and schedule reliability rather than simply filling every visible opening. It should specifically minimize avoidable first-call and midday reschedules while preserving capacity for real service demand.

10. Scheduled Days Off Feed Future Capacity

Known days off, vacations, training and approved reduced schedules must affect future booking capacity before the operating day begins. If a branch will have materially less qualified service capacity on a future day, Trades should reduce safely bookable capacity or recommend protective blocks in advance rather than discovering the shortage after the board is full.

11. Probabilistic Call Duration

A typical service-call booking window may be approximately two hours, with many calls extending toward four hours and lengthy repairs sometimes exceeding that. Trades should therefore model call duration as a probability distribution, not a fixed block.

The distribution should continuously recalibrate by call type, trade, equipment, age/condition, prior history, Van/technician era, access/difficulty, branch, geography, repair probability and actual completed outcomes.

Scheduled window != predicted completion time. The board should understand both.

12. First-Morning Call Protection

First calls should receive strong schedule-reliability weighting because an early overrun can cascade through the entire day. Trades should consider the probability that the first call completes with enough time to meet the next committed appointment and any employee availability boundary.

Where two otherwise appropriate assignments exist, the one producing materially higher probability of preserving the next commitment should receive scheduling preference unless urgency, qualification, safety or another governing factor dominates.

13. Midday Commitment Protection

The same principle applies to midday calls. Trades should identify calls immediately preceding hard customer commitments or employee windows and dynamically create only the buffer justified by the predicted overrun risk. Static empty gaps everywhere are unnecessary; risk-weighted buffers are preferred.

14. Reschedule-Risk Score

Field	Meaning
On-Time Probability	Probability the customer commitment can be met within the defined service window.
Reschedule Probability	Estimated probability the appointment must be materially moved/displaced.
Expected Start Range	Probability-informed arrival/start interval.
Expected Completion Range	Probability-informed completion interval.
Emergency Displacement Risk	Likelihood surge demand will consume the planned capacity.
Technician Overrun Risk	Likelihood prior work extends into this commitment.
Customer Flexibility	Policy/recorded flexibility, never invented.
Downstream Schedule Impact	Expected effect of delay on subsequent commitments and employee windows.

15. Protected Demand Reserve

Trades should not necessarily book every minute of every qualified Van. Branch/time-block emergency reserve should vary according to predicted urgent-call arrival rate, weather, season, day/time, staffing, skill mix and existing workload.

Total qualified capacity - protected commitments - employee restrictions - predicted overruns - emergency reserve = safely bookable capacity.

16. Demand Urgency Model

Class	Operational concept
Emergency	Immediate safety/property threat or other company-defined condition requiring the fastest appropriate response.
Urgent	Material service failure where delay has elevated customer/property/operational consequence.
Priority service	Important system failure requiring timely service but below emergency/urgent classification.
Routine service	Normal repair/service demand.
Maintenance / flexible	Planned work that can absorb more scheduling flexibility under authorized policy.

Urgency policy remains company-controlled. Trades learns the operational consequence of delay for each class but must not invent safety conditions or falsely escalate urgency.

17. Emergency Insertion Without Destroying the Board

An emergency does not automatically justify breaking multiple existing commitments. Trades evaluates whether the call can be absorbed through protected reserve, another qualified Van, nearby/cross-branch capacity, management resources, flexible maintenance movement or other authorized recovery options.

The engine should rank alternatives by response time, qualification, customer displacement, downstream schedule survival, labor/travel cost, overtime, contribution and operational risk, then preserve the chosen plan and outcome in receipts.

18. Morning Board Optimization

The morning board is a probabilistic best plan rather than a rigid schedule. It should: (1) protect high-confidence first commitments, (2) preserve enough flexibility for expected urgent demand, and (3) avoid creating a downstream cascade when early calls run long.

Chain A then continuously rebases the plan from actual real-world state as calls finish early, run long, employees become unavailable, emergencies arrive or capacity changes.

19. Demand x Day-Off x Booking Calibration

Trades should learn interactions rather than treating each variable independently. Examples include scheduled days off x temperature x weekday x skill mix x morning bookings; call-ins x emergency arrivals x maintenance load; and early-departure windows x expected repair duration x geography.

This allows future planning to answer not merely how many employees are scheduled, but whether the available skill/time distribution is sufficient for the expected demand curve while preserving existing customer commitments.

20. End-of-Day Settlement & Learning

Predicted / planned	Settled against
Call duration	Actual duration and reason for material variance.
Demand volume/mix	Actual calls by urgency, trade and call type.

Predicted / planned	Settled against
Emergency arrival probability	Actual emergency/urgent arrivals by time block.
Reschedule probability	Actual on-time, delayed, displaced or rescheduled result.
Protected reserve	Used capacity vs unused reserve and avoided displacement.
Staffing assumptions	Actual attendance, availability windows and skill capacity.
Morning trajectory	Actual route/day trajectory and rebase performance.
Recovery plan	Actual customer, labor, travel, overtime and economic consequence.

21. Chain D Integration

Availability and scheduling failures must be reconstructable through Chain D. If management approved a 4:00 PM return but the availability state never reached Dispatch, Chain D should identify the failed handoff rather than incorrectly labeling the dispatcher as the root cause.

22. Universal Analytics Integration

All workforce and scheduling variables become available to the v0.7 Universal Analytics layer. Human reviewers can generate reports, graphs, scales, probability analyses and models across days off, call-ins, demand, weather, urgency, booking blocks, first-call reliability, midday reliability, overtime, reschedules, cross-branch borrowing, labor/cost/profit and any other authorized metric.

23. Receipt Requirements

Material availability and capacity changes should record who entered/changed the constraint, constraint type, hard/soft state, when it became known, effective window, affected Van/role/branch, Trades recommendation, management action, booking/dispatch changes, customers affected, model/policy version, expected impact and final outcome.

24. Governing Objective

Serve the highest-urgency and highest-value legitimate demand while maximizing the probability that existing customer commitments and authorized employee availability windows are actually kept.

Trades should optimize usable capacity, not theoretical utilization. It should learn when leaving capacity open is valuable, when maintenance should remain bookable, when it should be blocked, and how much staffing/skill reserve a branch needs under different demand conditions.

25. Architecture Result

Future roster + scheduled days off + employee windows + historical demand + weather + current bookings + urgency mix + call-duration distributions + emergency-arrival probability -> Safe Bookable Capacity -> Morning Board -> Protected Capacity -> Live Rebase -> Emergency Insertion -> Reschedule Minimization -> End-of-Day Settlement -> Calibration.

End of Trades-Runtime Workforce Capacity, Availability & Demand-Aware Scheduling Addendum v0.8